Alkane

61. (b) Knocking - Sudden and irregular burning of the fuel mixture causing jerks against the piston and gives rise to violent sound. This is known as knocking.

62. (a) Alkanes — Petroleum mainly contains alkanes (along with some cycloalkanes), so it is primarily a mixture of saturated hydrocarbons.

63. (b) $n-$octane

- Boiling point depends on molecular mass. Greater the molecular mass higher will be the boiling point.
- Boiling point also depends on the structure. If two compounds have same molecular mass then straight chain or linear compound has higher boiling point.

64. (c) $\underset{\text{Potassium acetate}}{2CH_3COOK} + 2H_2O \xrightarrow{\text{Electrolysis}} CH_3-CH_3 + \underset{\text{Anode}}{2CO_2} + \underset{\text{Cathode}}{2KOH + H_2}$

65. (c) Ethane does not decolourise bromine solution because it is a saturated compound.

66. (b) $\underset{\text{Sodium acetate}}{CH_3COONa} + NaOH \xrightarrow{CaO} \underset{\text{Methane}}{CH_4} + Na_2CO_3$

67. (c) $CO + H_2$

68. (b) Octane number is related to the percentage of *iso*-octane since *iso*-octane is 81% hence octane number is 81%.

69. (d) All of these

70. (c) $2CH_3COOK + 2H_2O \xrightarrow{\text{Electrolysis}} \underbrace{2CO_2 + CH_3-CH_3 + 2KOH}_{\text{Anode}} + \underbrace{H_2}_{\text{Cathode}}$

71. (b) $CH_3-\underset{\underset{CH_3}{|}}{C}=CH_2 + HCl \rightarrow CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{Cl}{|}}{C}}-CH_3$

$CH \equiv CH + HCl \rightarrow CH_2=CH-Cl \xrightarrow{HCl} CH_3-CHCl_2$

72. (a) Boiling point of alkanes increases with the number of carbon atoms because surface area increases which increases the Vander Waal forces.

73. (c) High boiling constituents condense at the bottom of the column — Because the bottom of the fractionating column is hottest, high boiling fractions condense there first.

74. (c) The enthalpy of combustion *i.e.*, ΔH is always negative. It means combustion is an exothermic reaction.

75. **(b) — Gasoline refers to the light, volatile fraction of petroleum (petrol), not crude oil or residue.**

76. **(c)** — Cracking breaks high molecular weight hydrocarbons into lower molecular weight hydrocarbons.

77. **(c)** — n-Heptane has an octane number of 0 (iso-octane is assigned 100).

78. (c) $CH_3CH_2COONa + NaOH \xrightarrow{CaO} C_2H_6 + Na_2CO_3$

79. (a) $CH_3 - CH_2 - CH_2 - CH_3 + Br_2 \xrightarrow[130^oC]{\text{Light}} \underset{\substack{\text{2–Bromo butane} \\ \text{(Main product)}}}{CH_3 - \underset{\substack{| \\ Br}}{CH} - CH_2 - CH_3} + \underset{\substack{\text{1–Bromo butane} \\ \text{(Minor)}}}{CH_3 - CH_2 - CH_2 - CH_2 - Br}$

2-Bromobutane is the main product because

2° carbonium ion is more stable than 1°.

80. (c) $\underset{n\text{–Butane}}{CH_3CH_2CH_2CH_3} \xrightarrow{\text{Cracking}} CH_4 + CH_3 - CH = CH_2$